

Agri Intelligence Platform – Product Requirement Document

1. Project Overview

The goal is to build a smart agriculture platform that empowers farmers with real-time insights on crop health, soil condition, and financial credibility.

The platform will:

- Detect crop diseases using image analysis
- Monitor soil and vegetation health using geospatial data
- Generate a farm productivity-based credit profile
- Connect farmers with financial institutions and agri-product companies

This solution aims to create a data-driven agricultural ecosystem that improves yield, reduces risk, and enhances financial inclusion.

2. Core Objectives

- Improve crop health detection accuracy
- Enable data-driven irrigation decisions
- Digitize farm productivity records
- Provide credible data to banks & insurers
- Build a scalable platform for millions of farmers

3. Key Functional Modules

3.1 Crop Disease Detection ("Camera-Based Intelligence")

Description:

A feature that allows farmers to capture or upload images of crops/leaves to detect diseases.

Functional Requirements:

1. Upload or capture crop images via mobile device
2. Analyze image and identify:
 - Disease type (if any)
 - Severity level

Provide:

1. Disease name
2. Suggested treatment
3. Preventive measures

Expected Outcome:

Instant diagnosis of plant health

Reduction in manual inspection dependency

3.2 Soil & Crop Health Monitoring ("Geospatial Intelligence")

Description:

Use location-based analysis to monitor vegetation and soil moisture conditions.

Functional Requirements:

1. Farmer inputs or selects farm location (GPS-based)
2. System fetches and analyzes:
 - Vegetation health indicators
 - Soil moisture levels
3. Generate insights such as:
 - Irrigation recommendations
 - Crop stress alerts

Example Output:

“Soil moisture is low. Irrigation recommended within 24 hours.”

Expected Outcome:

Optimized water usage

Early detection of crop stress

3.3 Farm Credit Profiling ("Digital Farm Score")

Description:

Create a credibility score for farms based on productivity and historical data.

Functional Requirements:

1. Maintain farm-level data including:
 - Crop history
 - Health trends
 - Yield indicators (derived insights)
2. Generate a Farm Credit Score based on:
 - Land consistency
 - Crop health patterns
 - Productivity indicators

Expected Outcome:

Digitized, verifiable farm performance record

Reduced need for physical verification

3.4 Financial & Institutional Integration

Description:

Provide verified farm insights to banks and insurance providers.

Functional Requirements:

1. Generate farm reports for third-party institutions

Include:

- Farm health data
 - Risk indicators
 - Credit score
2. Enable secure sharing of reports with:
 - Banks
 - Insurance companies

Expected Outcome:

Faster loan approvals

Reduced risk for financial institutions

3.5 Advisory & Marketplace Integration

Description:

Provide actionable recommendations and enable product access.

Functional Requirements:

1. Suggest:

- Fertilizers
- Pesticides
- Crop care actions

2. Optional integration:

Purchase or connect with suppliers

Expected Outcome:

Increased farmer productivity

Monetization opportunities via partnerships

4. User Roles

1. Farmers

- Upload crop images
- View health insights
- Receive alerts and recommendations
- Access credit profile

2. Financial Institutions

- Access farm reports
- Evaluate creditworthiness

3. Admin / Platform Owner

- Monitor system performance
- Manage data and integrations
- Oversee partnerships

5. Business Model

Farmers

- Free access to platform features (to drive adoption)

Financial Institutions (Banks / Insurance)

- Pay-per-report model for farm verification
- Estimated pricing: ₹500 – ₹2,000 per report

Agri Companies

- Commission-based model
- Earn percentage on product recommendations and purchases

6. Key Features Summary

- Image-based disease detection
- Location-based soil and vegetation insights
- Automated irrigation recommendations
- Farm productivity scoring system
- Shareable farm reports for banks
- Advisory and product recommendation system

7. Scalability Expectations

- Support large number of farmers across regions
- Handle high volume of image and geospatial data
- Modular architecture for future integrations

8. Success Metrics

- Number of active farmers
- Accuracy of disease detection
- Reduction in crop loss
- Number of reports generated for institutions
- Conversion rate for financial services

9. Vision

To build a trusted digital infrastructure for agriculture, enabling:

- Smarter farming decisions
- Financial inclusion for farmers
- A connected ecosystem between farmers, banks, and agri-businesses

10. Immediate Next Step

The agency is expected to:

1. Understand the above requirements
2. Propose:
 - System architecture
 - Technology stack
 - Development roadmap
 - Cost estimation